EXP.NO:09

G-ARRAY SUM MAX PROBLEM

PROBLEM STATEMENT:

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

CODE:

```c
#include<stdio.h>
#include<stdlib.h>

int cmp(const void *a, const void *b){
    return *(int *)a - *(int *)b;
}

int main(){
    int n;
    scanf("%d", &n);
    int a[n];
    for(int i=0;i<n;i++)
        scanf("%d", &a[i]);

    qsort(a, n, sizeof(int), cmp);
    long long sum = 0;

    for(int i=0;i<n;i++)
        sum += (long long)a[i] * i;
    printf("%lld", sum);
    return 0;
}
```

EXP.NO:09

OUTPUT:



	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

RESULT:

The code is executed successfully.